

The goal of this homework is to determine the transverse displacement field and other characteristics of the simply supported plate, made of aluminum, shown in figure 1, assuming the plate model's basic hypotheses.

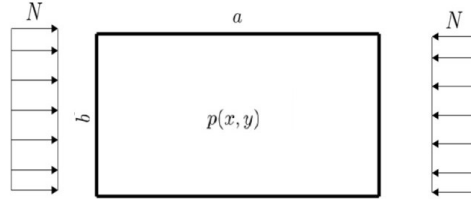


Figure 1: Analysed simply supported plate

DATA

- $h=3$ mm, thickness
- $a=2$ m
- $b=1$ m
- $E=72$ GPa, aluminum's Young modulus
- $\nu = 0.3$ aluminum's Poisson's modulus
- $\sigma_y = 350$ MPa aluminum's yielding stress

Point 1

Assumption: the plate is subject only to a transverse distributed load

$$p(x, y) = p_0 = 500 \text{ N/m}^2$$

Starting from the equation of motion in transverse direction for a simply supported plate under transverse loading:

$$D\nabla^4 w = p \quad (1)$$

where $D = \frac{Eh^3}{12(1-\nu^2)} = 178.02 \text{ Nm}$ is the bending stiffness of the plate and computing the boundary conditions in order to not admit bending moments:

$$\text{at } x=0, y=y \quad w = 0 \text{ and } M_x = 0 \longrightarrow \frac{\partial^2 w}{\partial y^2} \Big|_{x=0} = 0$$

$$\text{at } x=a, y=y \quad w = 0 \text{ and } M_x = 0 \longrightarrow \frac{\partial^2 w}{\partial y^2} \Big|_{x=a} = 0$$

$$\text{at } y=0, x=x \quad w = 0 \text{ and } M_y = 0 \longrightarrow \frac{\partial^2 w}{\partial x^2} \Big|_{y=0} = 0$$

$$\text{at } y=b, x=x \quad w = 0 \text{ and } M_y = 0 \longrightarrow \frac{\partial^2 w}{\partial x^2} \Big|_{y=b} = 0$$

it is possible to obtain the equation's solution, called also Navier solution, by assuming

to expand into a Fourier series both the solution and the transverse load:

$$w(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (2)$$

$$p(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} p_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (3)$$

The coefficient p_{mn} can be obtained by using the orthogonality property of the Fourier series terms, multiplying both terms of eq.3 by $\sin \frac{m'\pi x}{a} \sin \frac{n'\pi y}{b}$ with $m' = m$ and $n' = n$ and integrating on $0 \leq x \leq a$ and $0 \leq y \leq b$ so that it is obtained:

$$p_{mn} = \frac{4}{ab} \int_0^a \int_0^b p(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

This is the general expression for the coefficient p_{mn} but for this case, where $p(x, y)$ is a constant value equal to p_0 , it can be assumed as $\frac{16p_0}{\pi^2 mn}$.

Then, imposing the satisfaction of the field equation 1 one can obtain:

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left\{ a_{mn} \left[\left(\frac{m\pi}{a} \right)^4 + \left(\frac{n\pi}{b} \right)^4 + 2 \left(\frac{m\pi}{a} \right)^2 \left(\frac{n\pi}{b} \right)^2 \right] - \frac{p_{mn}}{D} \right\} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} = 0 \quad (4)$$

so, in order to solve the equation, which must be valid for each point in the plate, the term in curly brackets must be zero and the coefficient of the displacement's fourier expansion is:

$$a_{mn} = \frac{p_{mn}}{D\pi^4} \frac{1}{\left[\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2 \right]^2} \quad (5)$$

Substituting it in the expression for the transverse displacement, the final solution can be written as:

$$w(x, y) = \frac{1}{D\pi^4} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{16p_0}{\pi^2 mn \left[\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2 \right]^2} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (6)$$

In order to evaluate the maximum displacement value it is needed to decide which is the best approximation for the solution expressed in the eq.6 by evaluating the Fourier series up to a certain order. The following procedure for each solution's truncation order is to: evaluate the a_{mn} coefficients; evaluate the general expression for the transverse displacement and find the maximum value. In particular the maximum value is evaluated both graphically by looking at the plot of $w(x, y)$ and mathematically using the command *max* on *Matlab*. The best approximation will be decided computing the maximum difference between two consecutive truncated order Fourier series.

1st order truncation, up to m=1 and n=1

$$w_1(x, y) = a_{11} \sin \frac{\pi x}{a} \sin \frac{\pi y}{b} \quad \text{where} \quad a_{11} = 0.03 m$$

and the maximum value corresponds exactly to the central point of the plate i.e for $x = 1$ and $y = \frac{1}{2}$.

So $w_{max\ 1} = w(1, \frac{1}{2}) = 0.03\ m$.

3rd order truncation, up to m=3 and n=3

$$w_3(x, y) = a_{11} \sin \frac{\pi x}{a} \sin \frac{\pi y}{b} + a_{13} \sin \frac{\pi x}{a} \sin \frac{3\pi y}{b} + a_{31} \sin \frac{3\pi x}{a} \sin \frac{\pi y}{b} + a_{33} \sin \frac{3\pi x}{a} \sin \frac{3\pi y}{b}$$

where $a_{13} = 1.8266 * 10^{-4}\ m$, $a_{31} = 1.480 * 10^{-3}\ m$, $a_{33} = 4.116 * 10^{-5}\ m$.

As for the previous case, the maximum value is $w_{max\ 3} = 0.0284\ m$.

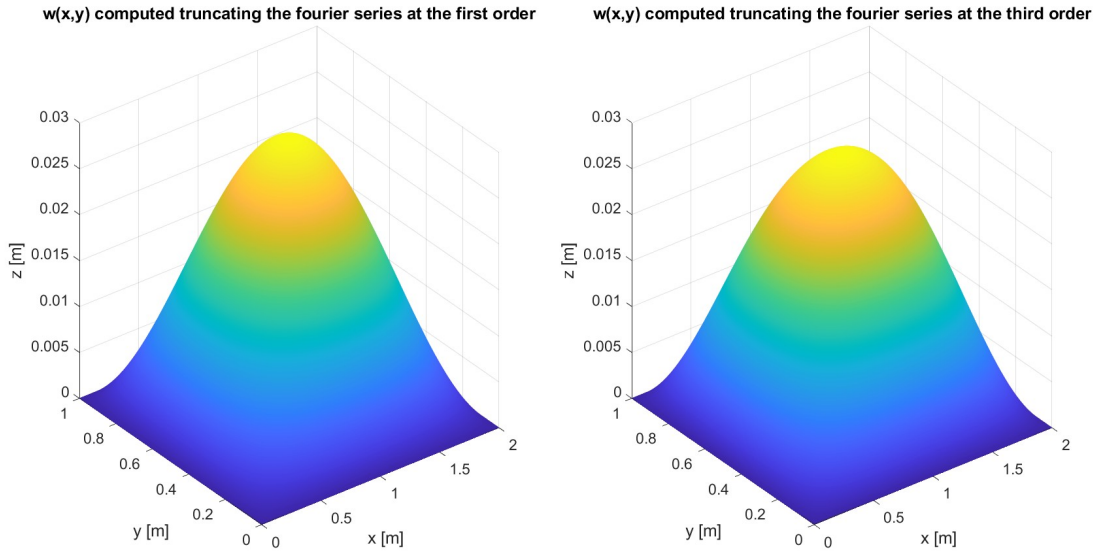


Figure 2: Transverse displacement field solution truncated to the 1st order on the left and 3rd order on the right for the solution of Point 1.

5th order truncation, up to m=5 and n=5

The expression for $w_5(x, y)$ is the same of the previous cases but adding the terms related to the new coefficients, that are: $a_{35} = 4.209 * 10^{-6}\ m$, $a_{53} = 1.344 * 10^{-5}$, $a_{55} = 1.920 * 10^{-6}$, $a_{15} = 1.471 * 10^{-5}$, $a_{51} = 1.784 * 10^{-4}$.

In this case the maximum value is $w_{max\ 5} = 0.0286\ m$.

7th order truncation, up to m=7 and n=7

The expression for $w_7(x, y)$ is the same of the previous cases but adding the terms related to the new coefficients a_{17} a_{71} a_{37} a_{73} a_{57} a_{75} a_{77} .

In this case the maximum value is $w_{max\ 7} = 0.0285\ m$.

The maximum difference between the transverse displacements of the solution truncated at the 3rd and 1st order is $1.3 \times 10^{-3} \text{ m}$, the maximum difference between the solution truncated at the 5th and 3rd order is 1.8×10^{-4} while the maximum difference between the 7th and 5th order is 3.5×10^{-5} . Obviously, as the order increases, the solution goes to convergence and the difference between the last two orders of truncation decreases. In this case, since the result in terms of maximum displacement is taken with an order of magnitude of 10^{-2} , the best choice in terms of approximation is the solution at the 5th order, which differs from the next approximation by 3 orders of magnitude less than the value of the maximum.

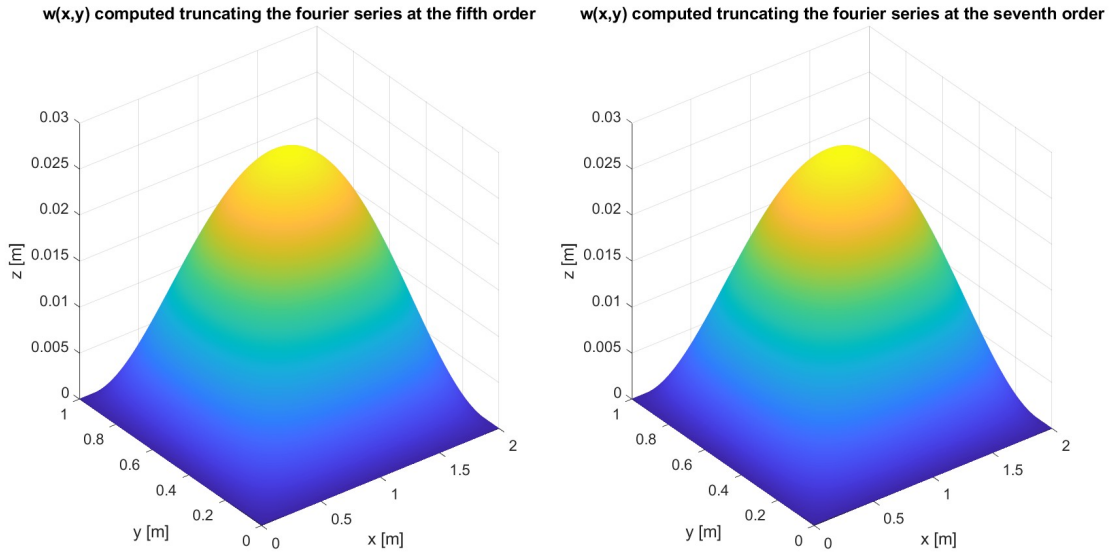


Figure 3: Transverse displacement field solution truncated to the 5th order on the left and 7th order on the right for the solution of point 1.

In fig. 2 and fig. 3 there are the plots in the 3D space of the transverse displacement field.

The solutions for the two cases considering or not the membrane-bending coupling are exactly the same because in this first point, since there is not any external load applied in the plane (x,y), the plate experiences only the bending behaviour.

Point 2

Assumptions: the plate is subject to both a transverse distributed load $p(x, y) = p_0 \sin \frac{\pi x}{a} \sin \frac{\pi y}{b}$, with $p_0 = 500 \text{ N/m}^2$ and a compressive load $N = 1000 \text{ N/m}$

1) No coupling between membrane and bending behaviour

This particular case takes into account only the transverse distributed load $p(x, y)$ so the starting equation of motion is the same as the previous case, eq.1. Also the boundary conditions are the same, but in this case there is not a constant load as the point one, so the coefficient p_{mn} of the Fourier expansion of the distributed load is, substituting $p(x, y)$ in the eq.4:

$$p_{mn} = \frac{4}{ab} \int_0^a \int_0^b p_0 \sin \frac{\pi x}{a} \sin \frac{\pi y}{b} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

The coefficient a_{mn} is evaluated as the same of eq.5 and the general solution is:

$$w(x, y) = \frac{1}{D\pi^4} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{p_{mn}}{[(\frac{m}{a})^2 + (\frac{n}{b})^2]^2} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (7)$$

In this case it is needed to evaluate all the p_{mn} coefficients for different m and n indices before evaluating a_{mn} :

$$p_{11} = \frac{4}{ab} \int_0^a \int_0^b p_0 \sin^2(\frac{\pi x}{a}) \sin^2(\frac{\pi y}{b}) dx dy = \frac{2p_0}{ab} = p_0$$

Applying the orthogonality property of the Fourier series terms, the only coefficient different from zero is that reported above while all the others are null. In this way, also $a_{11} = 0.01849$ is the only a_{mn} coefficient different from zero and the Fourier series can be expanded at most up to the first order.

The final expression for the transverse displacement is:

$$w(x, y) = a_{11} \sin \frac{\pi x}{a} \sin \frac{\pi y}{b} \quad (8)$$

and the maximum value is:

$$w_{\max}(1, 1/2) = a_{11} = 0.0185 \text{ m} \quad (9)$$

2) Coupling between membrane and bending behaviour

In this case they must be taken into account both the transverse distributed load $p(x, y)$ and the compressive load N , which is in the opposite direction of the x axis, so $N_x = -N$.

The general equation of motion is:

$$D\nabla^4 w = p(x, y) - N \frac{\partial^2 w}{\partial x^2} \quad (10)$$

and following the same method of the point 1, assuming the solution and the expression of $p(x, y)$ as an expansion of a Fourier series, the coefficient p_{mn} of the transverse load is the same of the previous case, while a_{mn} has an additional part at the denominator due to the presence of N :

$$a_{mn} = \frac{p_{mn}}{D\pi^4} \frac{1}{\left[\left(\frac{n}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2 - \frac{N}{D\pi^2}\left(\frac{m}{a}\right)^2}$$

At the end the final expression of the transverse displacement is:

$$w(x, y) = \frac{1}{D\pi^4} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{p_{mn}}{\left[\left(\frac{n}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2 - \frac{N}{D\pi^2}\left(\frac{m}{a}\right)^2} \sin \frac{m\pi x}{a} \sin \frac{\pi ny}{b}$$

which is reduced to the same eq.8 with a different $a_{11} = 0.0203 \text{ m}$.

So in this case the maximum value is:

$$w_{\max}(1, 1/2) = a_{11} = 0.0203 \text{ m} \quad (11)$$

This is a reasonable result because the transversal load for the coupled case is higher than the one of the uncoupled case, as it can be shown in fig.4. This is due to the fact that the coupling between the membrane behaviour, caused by the compressive load, determines an additional component along the z axis which is summed with the one determined by the bending behaviour.

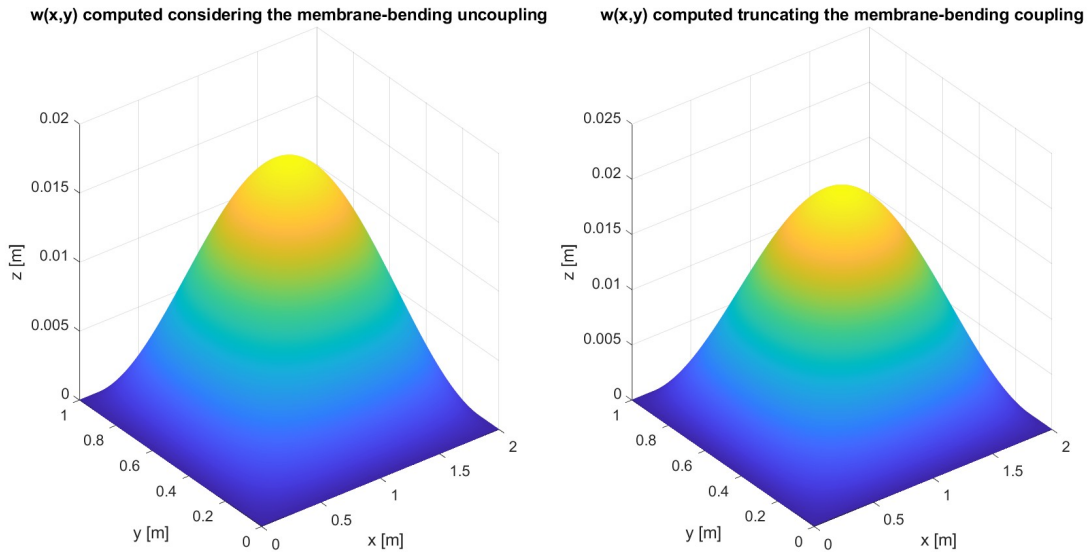


Figure 4: Transverse displacement field for the uncoupled case, on the left, and coupled case, on the right, for the solution of point 2.

Point 3

Assumption: the plate is subject only to a compressive load $N = 7200 \text{ N/m}$.
The general equation of motion is:

$$D\nabla^4 w = -N \frac{\partial^2 w}{\partial x^2} \quad (12)$$

with the same boundary conditions of point 1.

Assuming the transverse displacement as a fourier expansion and substituting it in the equilibrium equation it is obtained that:

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} \{ D\pi^4 [(\frac{m}{a})^2 + (\frac{n}{b})^2]^2 - N\pi^2 (\frac{m}{a})^2 \} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} = 0 \quad (13)$$

In order to obtain the expression for the critical compression load the transverse displacement must be infinite and it is possible only if its coefficient a_{mn} is infinite. In this case, solving the expression 13 for a_{mn} , it is obtained that it is infinite only if the term in curly brackets is zero (more precisely it must be a zero with an higher order than the zero on the right side of the equation, in order to avoid the indeterminate form).

So the expression for N is:

$$N = D\pi^2 \frac{[(\frac{m}{a})^2 + (\frac{n}{b})^2]^2}{(\frac{m}{a})^2} \quad (14)$$

The buckling critical load is find as the minimum value of N for which there is instability and that is only when n=1.

Now N_{cr} can be written in terms of m, a and b as:

$$N_{cr} = \frac{D\pi^2}{b^2} K \quad \text{where} \quad K = [\frac{m}{a}b + \frac{a}{mb}]^2$$

The minimum value of N_{cr} is for K=4 and it corresponds to the value m=2.

So $N_{cr} = 7021 \text{ N/m}$ and it is less then the applied compressive load $N = 7200 \text{ N/m}$. This means that, in order to avoid buckling, it is necessary to change the size of the plate. In particular, since N_{cr} must increase, it is possible to increase the thickness h of the plate in order to increase D and obtain a reasonable value of the MOS considering the safety factor n=1.5.

In a more general way it must be that $N_{cr} > nN$ and it is true only if $h > 3.46 \text{ mm}$.

In order to choose the right h the MOS is evaluated as

$$MOS = \frac{N_{cr}}{FOS * N} - 1$$

and with the given thickness of 3mm $MOS = -0.35$ which, as expected, it is a not admissible value.

Figure 5 shows the evolution of the MOS starting from the initial thickness value of 3 mm and it can be seen that it is necessary at least a value of $h = 3.46\text{ mm}$ to have a $MOS > 0$, which is the general condition obtained at the beginning. Increasing the plate's thickness at $h=3.6\text{ mm}$ it is obtained a reasonable $MOS = 12.5\%$ and $N_{cr} = 12144\text{ N/m}$ which is greater than N .

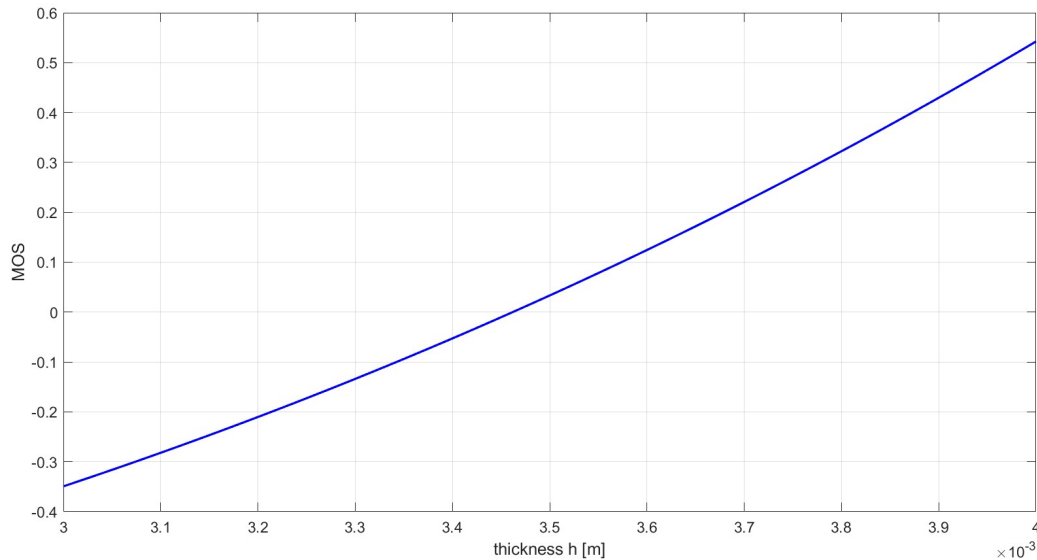


Figure 5: Evolution of MOS as a function of the thickness.